

Adapting the Harrow-Hassidim-Lloyd Algorithm to Quantum Many-Body Theory: Paper Review

Shivaji Roy Chowdhury: EP23B049
Department of Engineering Physics
Indian Institute of Technology, Madras

A review and implementation study of the paper “Adapting the Harrow-Hassidim-Lloyd Algorithm to Quantum Many-Body Theory” by Nishanth Baskaran *et al.*, published in *Physical Review Research* **5**, 043113 (2023), is presented. DOI: 10.1103/PhysRevResearch.5.043113. A connection between the Harrow-Hassidim-Lloyd (HHL) quantum linear systems algorithm and the linearized coupled cluster (LCC) equations of quantum many-body theory is established in the original work, allowing electronic structure problems to be reformulated as systems of linear equations amenable to quantum computation. Although the standard HHL algorithm provides a potential exponential quantum advantage, its practical realization on noisy intermediate-scale quantum (NISQ) devices is limited by large circuit depths and extensive controlled operations. To address these challenges, a family of optimized algorithms, namely **HHLite** and **AdaptHHLite**, is proposed, to achieve two levels of circuit compression. In the present report, the theoretical foundations of these algorithms are reviewed and implementations are carried out using Qiskit on noiseless simulators, noisy simulators, and IBM quantum hardware. Numerical results for the H₂ and LiH molecular systems are presented with NISQ-era computation constraints,

I. PROBLEM SETUP

A. Chemistry-Inspired Linear Equations

In quantum chemistry, the accurate determination of electronic ground-state energies is fundamental to the study of molecular structure and dynamics. Within the Born-Oppenheimer approximation, the electronic structure problem is reduced to solving the time-independent Schrödinger equation

$$\hat{H}|\Psi\rangle = E|\Psi\rangle,$$

where $\hat{H}$ is the electronic Hamiltonian, $|\Psi\rangle$ is the many-electron wavefunction, and E is the corresponding energy eigenvalue. For many-electron systems, the exact wavefunction cannot generally be obtained efficiently due to the exponential growth of the Hilbert space dimension with system size, and approximate many-body methods are therefore employed.

One of the most successful approaches is the coupled cluster (CC) method, in which the correlated wavefunction is represented through the exponential ansatz

$$|\Psi\rangle = e^{\hat{T}}|\Phi_0\rangle, \quad (1)$$

where $|\Phi_0\rangle$ denotes the Hartree-Fock reference determinant and

$$\hat{T} = \hat{T}_1 + \hat{T}_2 + \cdots + \hat{T}_N$$

is the cluster excitation operator for an N -electron system. Here, $\hat{T}_1$ and $\hat{T}_2$ correspond to the single and double excitation operators respectively. It is important to note that the operator $e^{\hat{T}}$ is not unitary.

In the spin-free formulation, the excitation operators are written as

$$\hat{T}_1 = \sum_{ia} t_i^a \{\hat{e}_i^a\},$$

and

$$\hat{T}_2 = \sum_{ijab} t_{ij}^{ab} \{\hat{e}_{ij}^{ab}\},$$

where indices $i, j, \dots$ denote occupied (hole) orbitals, indices $a, b, \dots$ denote unoccupied (particle) orbitals, and t_i^a, t_{ij}^{ab} are the corresponding cluster amplitudes. The operators $\hat{e}$ generate orbital substitutions out of the Hartree-Fock reference state.

Although the full coupled cluster formalism provides highly accurate results, the resulting equations are non-linear and computationally demanding. In the present work, the linearised coupled cluster (LCC) approximation is considered, in which only terms linear in the excitation operator $\hat{T}$ are retained. A detailed derivation is provided in Appendix A; only the essential steps relevant to the present discussion are summarised here.

The exponential operator is first linearised according to

$$e^{\hat{T}} \approx 1 + \hat{T}.$$

Under this approximation, the coupled cluster wavefunction becomes

$$|\Psi_{\text{LCC}}\rangle \approx (1 + \hat{T})|\Phi_0\rangle.$$

Substituting this expression into the Schrödinger equation gives

$$\hat{H}(1 + \hat{T})|\Phi_0\rangle = E(1 + \hat{T})|\Phi_0\rangle.$$

Expanding both sides yields

$$\hat{H}|\Phi_0\rangle + \hat{H}\hat{T}|\Phi_0\rangle = E|\Phi_0\rangle + E\hat{T}|\Phi_0\rangle.$$

The excitation operator may be expanded in the excitation basis as

$$\hat{T} = \sum_Q t_Q \{\hat{e}_Q\},$$

where t_Q denotes the Q th cluster amplitude. Projection onto the excited determinants

$$|\chi_P\rangle = \{\hat{e}_P\}|\Phi_0\rangle$$

then gives the LCC amplitude equations

$$\langle\chi_P|\hat{H}|\Phi_0\rangle + \sum_Q \langle\chi_P|\hat{H}|\chi_Q\rangle t_Q = 0, \quad \forall P.$$

Rearranging gives

$$\sum_Q \langle\chi_P|\hat{H}|\chi_Q\rangle t_Q = -\langle\chi_P|\hat{H}|\Phi_0\rangle, \quad \forall P.$$

The above expression is recognised as a system of linear equations of the form

$$A\mathbf{x} = -\mathbf{b}, \quad (2)$$

where the matrix elements are identified as

$$A_{PQ} = \langle\chi_P|\hat{H}|\chi_Q\rangle,$$

the vector elements as

$$b_P = \langle\chi_P|\hat{H}|\Phi_0\rangle,$$

and the vector $\mathbf{x}$ contains the unknown cluster amplitudes t_Q .

The determination of the LCC amplitudes is therefore reduced to solving a linear system, thereby enabling the direct application of quantum linear systems algorithms such as the Harrow–Hassidim–Lloyd (HHL) algorithm.

Once the amplitudes have been obtained, the LCC correlation energy is given by

$$E_{\text{corr}} = \langle\Phi_0|\hat{H}|\chi_P\rangle t_P.$$

Using

$$\mathbf{x} = -A^{-1}\mathbf{b},$$

the correlation energy may equivalently be written as

$$E_{\text{corr}} = -\mathbf{b}^\dagger A^{-1}\mathbf{b}. \quad (3)$$

Physically, E_{corr} represents the correction to the Hartree–Fock mean-field energy arising from correlated electron motion. While the Hartree–Fock approximation accounts for the average interaction experienced by each electron, dynamical electron–electron correlations are neglected at the mean-field level. The quantity E_{corr} therefore measures the additional stabilisation introduced by many-body quantum correlations and serves as an important indicator of the accuracy of an electronic structure method.

B. Hong–Ou–Mandel Module

Before proceeding to the modifications introduced in HHLite and AdaptHHLite, it is worth discussing the Hong–Ou–Mandel (HOM) module employed by the authors. This module is used to estimate the overlap between the states $|b\rangle$ and $|x\rangle$, where

$$|x\rangle \propto A^{-1}|b\rangle,$$

thereby enabling the evaluation of the correlation energy

$$E_{\text{corr}} = -\mathbf{b}^\dagger A^{-1}\mathbf{b}. \quad (4)$$

The original Hong–Ou–Mandel effect arises in quantum optics when two indistinguishable photons enter a balanced beam splitter from different input ports. Due to quantum interference, the probability amplitudes corresponding to the photons exiting through different output ports interfere destructively, resulting in the suppression of coincidence counts. The visibility of this interference pattern depends directly on the distinguishability, and therefore the overlap, of the photon states. An illustration of the HOM effect is shown in Fig. 1.

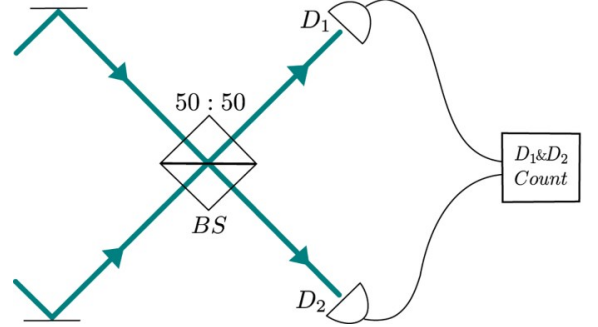


FIG. 1. Schematic illustration of the Hong–Ou–Mandel effect. Two indistinguishable photons incident on a balanced beam splitter undergo destructive interference in the coincidence channel, resulting in photon bunching. The interference visibility depends on the overlap between the incoming quantum states.

The HOM module employed in AdaptHHL is inspired by this interference-based overlap estimation procedure. In the present setting, physical photons and beam splitters are replaced by qubit registers and Hadamard operations respectively, while the overlap between quantum states is extracted from measurement statistics generated through quantum interference.

The complete module used in the AdaptHHL framework is shown in Fig. 2. The left portion corresponds to the standard HHL circuit, in which quantum phase estimation (QPE), controlled rotations, and inverse QPE are used.

The right portion corresponds to the HOM-inspired overlap estimation module. Here, interference between the states $|b\rangle$ and $|x\rangle$ is generated through controlled operations and Hadamard transformations, allowing the

overlap $\langle b|x \rangle$ to be reconstructed from measurement probabilities.

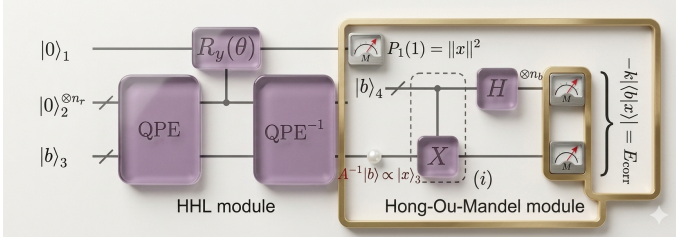


FIG. 2. The HHL and Hong–Ou–Mandel modules employed in the AdaptHHL framework. The interference between the states $|b\rangle$ and $|x\rangle$ is used to evaluate the overlap $\langle b|x \rangle$, and consequently the correlation energy.

For an n -qubit system, the overlap between the two states is obtained from the relation

$$|\langle b|x \rangle|^2 = \sum_{\alpha, \beta \in \{0,1\}^n} (-1)^{\alpha \cdot \beta} P(\alpha\beta), \quad (5)$$

where α and β denote measured n -bit strings, $\alpha \cdot \beta$ represents the bitwise inner product modulo two, and $P(\alpha\beta)$ is the joint probability associated with obtaining the corresponding bit strings upon measurement.

The factor $(-1)^{\alpha \cdot \beta}$ introduces constructive and destructive interference between computational paths in a manner analogous to optical HOM interference. Through this parity-weighted summation, all contributions except the desired overlap term cancel, allowing the quantity $|\langle b|x \rangle|^2$ to be extracted directly from measurement statistics.

A detailed derivation of the above overlap formula and its connection to interference generated by Hadamard transformations is provided in Appendix B.

C. Standard HHL Algorithm and Its Limitations

The Harrow–Hassidim–Lloyd (HHL) algorithm is a quantum linear systems algorithm designed to prepare a quantum state proportional to the solution of a linear system

$$A\mathbf{x} = \mathbf{b},$$

where A is assumed to be Hermitian, sparse, and well-conditioned. Rather than explicitly outputting the vector $\mathbf{x}$, the algorithm prepares the normalized quantum state

$$|x\rangle \propto A^{-1}|b\rangle.$$

The input vector $\mathbf{b}$ is encoded into a quantum state

$$|b\rangle = \sum_j \beta_j |u_j\rangle,$$

where $\{|u_j\rangle\}$ are the eigenvectors of A satisfying

$$A|u_j\rangle = \lambda_j |u_j\rangle.$$

The principal stages of the HHL algorithm are summarised below:

- **State Preparation:** The normalized right-hand-side vector $\mathbf{b}$ is encoded into the quantum state $|b\rangle$.
- **Quantum Phase Estimation (QPE):** Controlled time-evolution operators based on

$$U = e^{iAt}$$

are applied to estimate the eigenvalues λ_j of the matrix A . After QPE, the joint state becomes

$$\sum_j \beta_j |\tilde{\lambda}_j\rangle |u_j\rangle,$$

where $|\tilde{\lambda}_j\rangle$ stores the binary approximation of the eigenvalue.

- **Controlled Rotation:** An ancilla qubit is rotated according to the estimated eigenvalues such that inverse eigenvalue information is encoded into the ancilla amplitudes. The resulting state may be written as

$$\sum_j \beta_j \left(\sqrt{1 - \frac{C^2}{\tilde{\lambda}_j^2}} |0\rangle + \frac{C}{\tilde{\lambda}_j} |1\rangle \right) |\tilde{\lambda}_j\rangle |u_j\rangle,$$

where C is a normalization constant chosen to ensure valid rotation angles.

- **Inverse QPE:** The phase estimation procedure is reversed in order to uncompute the clock register and disentangle it from the system register. After the inverse QPE stage, the state becomes

$$\sum_j \beta_j \left(\sqrt{1 - \frac{C^2}{\tilde{\lambda}_j^2}} |0\rangle + \frac{C}{\tilde{\lambda}_j} |1\rangle \right) |0\rangle |u_j\rangle.$$

- **Postselection:** Measuring the ancilla qubit in the state $|1\rangle$ leaves the system register in the state

$$|x\rangle \propto \sum_j \frac{\beta_j}{\lambda_j} |u_j\rangle = A^{-1}|b\rangle.$$

A representative implementation of the standard HHL circuit used in the present work is shown in Fig. 3. The circuit contains the QPE stage, controlled powers of the unitary evolution operator, inverse QFT operations, and the large collection of multi-controlled rotation gates required to implement the inverse eigenvalue transformation.

Although HHL provides an elegant quantum algorithm for solving linear systems, practical implementations on present-day quantum hardware remain challenging due to several important limitations.

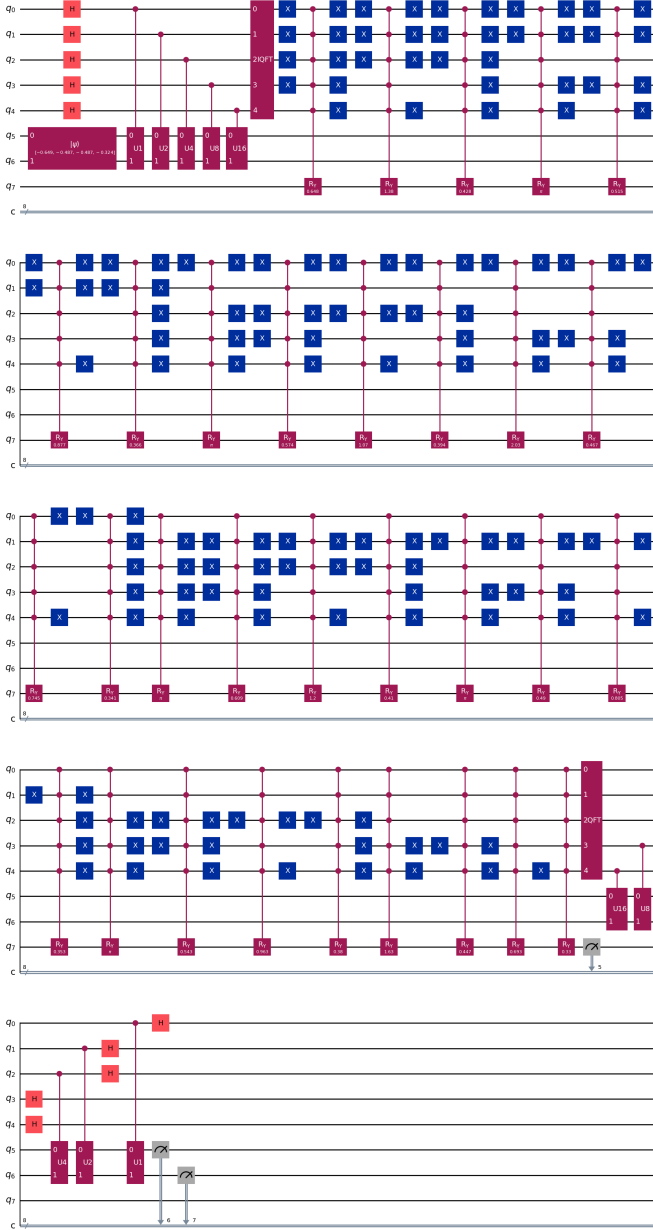


FIG. 3. Circuit implementation of the standard HHL algorithm, using 5 clock qubits. The resulting circuit depth and gate complexity render direct implementation difficult on present-day NISQ hardware.

The first major difficulty arises from the quantum phase estimation stage itself. Accurate eigenvalue estimation requires multiple clock qubits together with controlled applications of

$$U^{2^k} = e^{iA2^k t},$$

leading to substantial circuit depth. Since the controlled unitary operators must remain coherent throughout the computation, the QPE stage becomes highly susceptible to decoherence and gate noise on NISQ-era devices.

A second and more severe bottleneck arises from the controlled rotation stage. The inverse eigenvalue transformation requires different ancilla rotations conditioned on the eigenvalue register. In practice, this leads to a large number of multi-controlled R_y gates whose number grows rapidly with clock precision. As seen in Fig. 3, the controlled rotation module dominates the overall circuit complexity and introduces a substantial number of entangling operations.

A further limitation concerns the scaling of the Hamiltonian and the choice of evolution time t . The eigenvalues must be appropriately mapped into the resolvable phase range of QPE. Poor scaling may lead to overlapping eigenvalue phases and inaccurate inversion, while larger evolution times generally increase circuit depth and noise accumulation.

However, in many quantum chemistry applications, the complete reconstruction of the state $|x\rangle$ is not required explicitly. Instead, quantities such as the overlap

$$\langle b|x\rangle,$$

from which the correlation energy may be extracted, are often sufficient. This observation enables significant simplifications of the standard HHL procedure. By exploiting the structure of the overlap estimation problem, reduced-depth variants such as HHLite and AdaptHHLite may be constructed while still retaining remarkably accurate estimates of the correlation energy.

II. PROPOSED OPTIMIZATIONS AND IMPLEMENTATION

A. Molecular Systems: LiH and H₂

The performance of algorithms was investigated using reduced linear systems derived from the linearised coupled cluster treatment of the hydrogen molecule (H₂) and lithium hydride (LiH).

1. Hydrogen Molecule (H₂)

For the H₂ system, the reduced matrix representation was given by

$$A_{H_2} = \begin{pmatrix} 1.12854 & 0 & 0 & 0.03593 \\ 0 & 1.44616 & 0.08368 & 0 \\ 0 & 0.08368 & 1.44616 & 0 \\ 0.03593 & 0 & 0 & 1.94607 \end{pmatrix}$$

$$\mathbf{b}_{H_2}^T = (-0.2 \ -0.15 \ -0.15 \ -0.1).$$

The eigenvalue spectrum of A_{H_2} was found to be

$$\{1.126964, 1.362480, 1.529840, 1.947646\},$$

with condition number

$$\kappa(A_{H_2}) = 1.7282.$$

The corresponding classical correlation energy was

$$E_{\text{corr}}^{(\text{H}_2)} = -0.06937.$$

2. Lithium Hydride (LiH)

For the LiH system, the reduced matrix and vector were

$$A_{\text{LiH}} = \begin{pmatrix} 0.500000 & -0.051643 & 0.051643 & 0.011732 \\ -0.051643 & 0.648716 & -0.011732 & 0 \\ 0.051643 & -0.011732 & 0.662297 & 0 \\ 0.011732 & 0 & 0 & 1.211248 \end{pmatrix}$$

$$\mathbf{b}_{\text{LiH}}^T = (-0.2 \ -0.15 \ -0.15 \ -0.1)$$

The eigenvalues of A_{LiH} were found to be

$$\{0.472399, 0.643024, 0.695393, 1.211444\},$$

corresponding to a condition number

$$\kappa(A_{\text{LiH}}) = 2.5644.$$

The corresponding classical correlation energy was

$$E_{\text{corr}}^{(\text{LiH})} = -0.15890.$$

Both systems possess relatively small condition numbers making them suitable benchmark systems for investigating HHL-based algorithms.

B. The Lite Module

The HHLite (*HHL implementation using truncated eigenspace*) algorithm is a reduced-depth variant of the standard HHL algorithm. The central idea underlying HHLite is the *multi-qubit fixing* procedure, in which selected qubits of the quantum phase estimation (QPE) register are fixed to classical values based on their measured probability distributions.

The HHLite procedure may be summarised as follows:

- **Initial QPE execution:**

A standard quantum phase estimation circuit is first executed in order to obtain the probability distribution associated with the clock register qubits.

- **Evaluation of qubit probabilities:**

For each clock qubit i , the probability

$$P_i(1)$$

of measuring the qubit in the state $|1\rangle$ is computed from repeated measurements of the QPE circuit.

- **Threshold-based qubit fixing:**

A probability threshold θ is introduced. If

$$P_i(1) > \theta,$$

the qubit is fixed to the classical state $|1\rangle$, while if

$$P_i(1) < 1 - \theta,$$

the qubit is fixed to the state $|0\rangle$. Qubits whose probabilities lie within the intermediate region are retained as active quantum qubits.

- **Reduced controlled evolution:**

Controlled unitary operations associated with fixed qubits are simplified. Since these qubits no longer exist in coherent superposition, the corresponding controlled operations may either be replaced by unconditional unitary evolutions or removed entirely.

The controlled R_y rotation stage is constructed only over the remaining active clock qubits.

- **Inverse QPE and overlap estimation:**

The inverse QPE stage is applied over the remaining active qubits, following which the overlap $\langle b|x\rangle$ is estimated using the Hong–Ou–Mandel module discussed previously.

The principal advantage of HHLite lies in the reduction of controlled unitary operations appearing within the quantum phase estimation stage. In standard HHL, the implementation of controlled evolutions of the form

$$U^{2^k} = e^{iA2^k t}$$

constitutes one of the dominant sources of circuit depth and entangling gate overhead. Through the multi-qubit fixing procedure, several clock qubits are removed from coherent superposition, thereby eliminating the need for their corresponding controlled unitary implementations. As a result, the depth of both the forward and inverse QPE stages is substantially reduced, leading to significantly shallower circuits while still preserving accurate estimates of the correlation energy.

1. Results for H_2

The HHLite algorithm was first benchmarked using the reduced H_2 molecular Hamiltonian introduced previously. An initial quantum phase estimation (QPE) run was performed over 2048 shots in order to determine the probability distribution associated with the clock register qubits. The measured probabilities for obtaining the state $|1\rangle$ on each clock qubit were found to be

$$P(1) = [0.830, 0.583, 0.857, 0.406, 0.522].$$

Using the threshold-based multi-qubit fixing procedure described earlier, the corresponding fixing order was determined to be

$$[2, 0, 3, 1, 4].$$

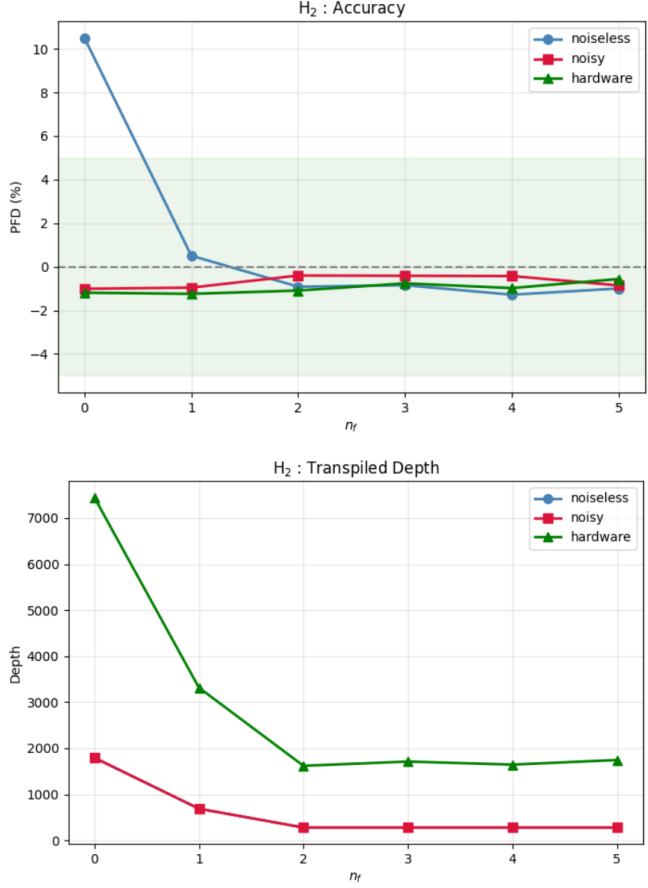


FIG. 4. HHLite performance for the H_2 molecular system. **Top:** Percentage fractional deviation (PFD) of the estimated correlation energy from the classical value as a function of the number of attempted fixed clock qubits n_f . **Bottom:** Corresponding transpiled circuit depth for noiseless simulation, noisy simulation, and IBM hardware execution.

This indicates that the third and first clock qubits exhibited the strongest deterministic bias and were therefore selected first during the qubit-fixing procedure.

The HHLite algorithm was subsequently executed under the following three settings:

1. **noiseless** QASM simulation,
2. simulation in the presence of **depolarising noise**,
3. execution on **IBM Quantum hardware** using the `ibm_fez` backend.

The complete numerical results for all three implementations are provided in Appendix C 1.

Several important trends may be observed from Fig. 4:

- **Depth reduction:**

A substantial reduction in **transpiled circuit depth** was achieved after fixing only two clock qubits. In the noiseless implementation, the depth decreased from approximately 1800 to **below 300**.

- **Accuracy retention:**

Despite the reduced circuit complexity, the estimated correlation energy remained within approximately 1% of the classical value across most implementations. The PFD mentioned is calculated as

$$\text{PFD} = \frac{E_{\text{est}} - E_{\text{classical}}}{|E_{\text{classical}}|} \times 100 \quad (6)$$

- **CX vs CZ gate counts:**

For noiseless and noisy simulations, entangling gate counts were quantified using the number of **CX** gates appearing in the transpiled circuit. For IBM hardware executions, the number of **CZ** gates was reported instead, since CZ constitutes the native two-qubit entangling gate of the `ibm_fez` backend.

- **Transpiled depth:**

All reported circuit depths correspond to *transpiled* circuit depths rather than logical circuit depths. Consequently, the hardware depths additionally include routing, basis decomposition, and hardware-native compilation overheads.

- **Runtime:**

The reported runtime values correspond to local execution and transpilation times measured during simulation or hardware submission. For IBM hardware executions, these values do not represent the true physical quantum execution duration, since backend queue delays and internal scheduling information are not directly accessible through the public runtime interface.

- **Saturation beyond $n_f = 2$:**

No further reduction in circuit depth was observed beyond $n_f = 2$. Although additional qubits were considered during the fixing procedure, only the first two clock qubits satisfied the prescribed threshold conditions. The remaining qubits possessed probabilities too close to 0.5 and therefore continued to be treated as active quantum qubits. Consequently, the effective number of fixed qubits remained unchanged for $n_f > 2$.

2. Results for LiH

The HHLite algorithm was next benchmarked using the reduced LiH molecular Hamiltonian. An initial QPE run was performed over 2048 shots in order to determine the probability distribution associated with the clock register qubits. The measured probabilities for obtaining the state $|1\rangle$ on each clock qubit were found to be

$$P(1) = [0.261, 0.244, 0.526, 0.522, 0.588].$$

The corresponding fixing order obtained from the multi-qubit fixing procedure was

$$[1, 0, 4, 2, 3].$$

Unlike the H_2 system, however, none of the clock qubits satisfied the prescribed threshold conditions required for deterministic fixing. Although several qubits exhibited mild biases away from 0.5, the probabilities remained insufficiently close to either 0 or 1 to permit reliable qubit fixing.

Consequently, the effective number of fixed qubits remained zero throughout the entire HHLite sweep, even as larger values of n_f were considered. As a result, essentially no reduction in circuit depth or entangling gate count was observed across the noiseless, noisy, or hardware implementations.

The complete numerical results for all LiH HHLite implementations are provided in Appendix C 2.

This behaviour highlights an important limitation of the HHLite approach. The effectiveness of the multi-qubit fixing procedure depends strongly upon the probability distribution generated by the QPE register. If the eigenvalue distribution produces clock qubits with probabilities clustered near 0.5, the fixing procedure becomes ineffective and little or no circuit simplification may be achieved.

The LiH system therefore provides a natural motivation for the AdaptHHLite framework introduced in the next section. Rather than relying directly on the raw eigenvalue spectrum appearing in standard HHL, AdaptHHLite modifies the Hamiltonian scaling procedure in order to produce a more favourable distribution of QPE probabilities, thereby enabling substantially more effective qubit fixing and circuit compression.

C. The AdaptHHLite Algorithm

As observed for the LiH system, the success of the HHLite protocol depends strongly upon the probability distribution generated within the quantum phase estimation (QPE) register. The AdaptHHLite framework was introduced in order to address this limitation. Instead of directly applying HHL to the original matrix A , the matrix is first rescaled in such a way that its eigenvalues become more favourably distributed for representation within the QPE register.

The matrix is first rescaled according to

$$A_{sc} = sA,$$

where the scaling parameter is chosen as

$$s = \frac{2^{-n_r}}{\tilde{d}_{\min}},$$

with n_r denoting the number of QPE register qubits and $\tilde{d}_{\min}$ representing an estimate of the smallest eigenvalue of A .

To motivate this choice, consider a Hermitian matrix of the form

$$A = \begin{bmatrix} d_{\max} & d_{12} \\ d_{21} & d_{\min} \end{bmatrix},$$

where $d_{\max}$ and $d_{\min}$ denote the largest and smallest diagonal entries respectively.

The scaling factor is chosen so that the eigenvalues of the scaled matrix may be represented efficiently within the finite precision of the QPE register. The key requirements may be summarised as follows:

- **Eigenvalues must remain below unity:**

The scaled matrix should satisfy

$$2^{-n_r} \frac{d_{\max}}{\tilde{d}_{\min}} < 1,$$

ensuring that the largest eigenvalue of the scaled matrix remains within the range resolvable by the QPE register.

- **Small eigenvalues must remain representable:**

The scaling must also ensure that the smallest eigenvalue is not pushed below the resolution of the register. This leads to the condition

$$2^{-n_r} \frac{d_{\min}}{\tilde{d}_{\min}} \geq 2^{-n_r}.$$

- **Constraint on the scaling estimate:**

Combining the above conditions gives

$$d_{\min} \geq \tilde{d}_{\min} > 2^{-n_r} d_{\max}.$$

The estimate $\tilde{d}_{\min}$ must therefore be chosen carefully so that all relevant eigenvalues remain appropriately represented within the QPE spectrum. For our cases of smaller matrices of dimensions 4×4 , *it suffices to use the smallest diagonal element as an estimate of the smallest eigenvalue*, instead of constructing a classical algorithm for doing the same.

- **Improved QPE probability distributions:**

After rescaling, the eigenvalues become concentrated near binary fractions naturally captured by the QPE register. Consequently, the resulting clock-qubit probabilities become significantly more biased toward deterministic values close to either 0 or 1. The improved probability distribution substantially enhances the effectiveness of the HHLite multi-qubit fixing procedure. A larger number of clock qubits may therefore be fixed reliably. A significant fraction of the controlled unitary evolutions

$$U^{2^k}$$

appearing within the QPE stage may be simplified or removed entirely, leading to major reductions in circuit depth and entangling gate count.

- **Simplified controlled rotations:**

The rescaling procedure additionally simplifies the controlled rotation stage appearing within HHL. In standard HHL, the ancilla rotation angles are given by

$$\theta_i = 2 \arcsin\left(\frac{c}{\tilde{\lambda}_i}\right),$$

where $\tilde{\lambda}_i$ denotes the eigenvalue encoded by QPE and c is a normalisation constant constrained by the smallest eigenvalue.

In the AdaptHHLite framework, the scaling is chosen such that

$$\tilde{\lambda}_i \approx \frac{i}{2^{n_r}},$$

allowing the choice $c = 2^{-n_r}$.

Consequently, $\frac{c}{\tilde{\lambda}_i} = \frac{1}{i}$

and the controlled rotation angles assume simple fixed values determined only by the integer index i . This significantly reduces the complexity of the controlled rotation network and further contributes to the overall circuit compression achieved within AdaptHHLite.

As will be demonstrated in the following section, this modification enables substantial circuit compression even for systems such as LiH, where the original HHLite protocol fails to produce meaningful qubit fixing.

1. AdaptHHLite Results for H_2

The AdaptHHLite framework was first benchmarked using the reduced H_2 molecular Hamiltonian. The scaling parameters obtained for the system were found to be:

- **Minimum diagonal estimate:** $d_{\min} = 1.128540$
- **Scaling factor:** $s = 0.027691$
- **Controlled rotation constant:** $C = 0.031250$
- **Scaled eigenvalues:**
[0.03121, 0.05393, 0.04236, 0.03773]
- **QPE clock-qubit probabilities:**
 $P(1) = [0.106, 0.053, 0.052, 0.048, 0.048]$
- **Qubit fixing order:** [4, 3, 2, 1, 0]

The rescaling procedure produces a dramatically more deterministic probability distribution compared to standard HHLite. All clock-qubit probabilities become strongly biased toward the state $|0\rangle$, allowing substantially more effective qubit fixing throughout the AdaptHHLite protocol.

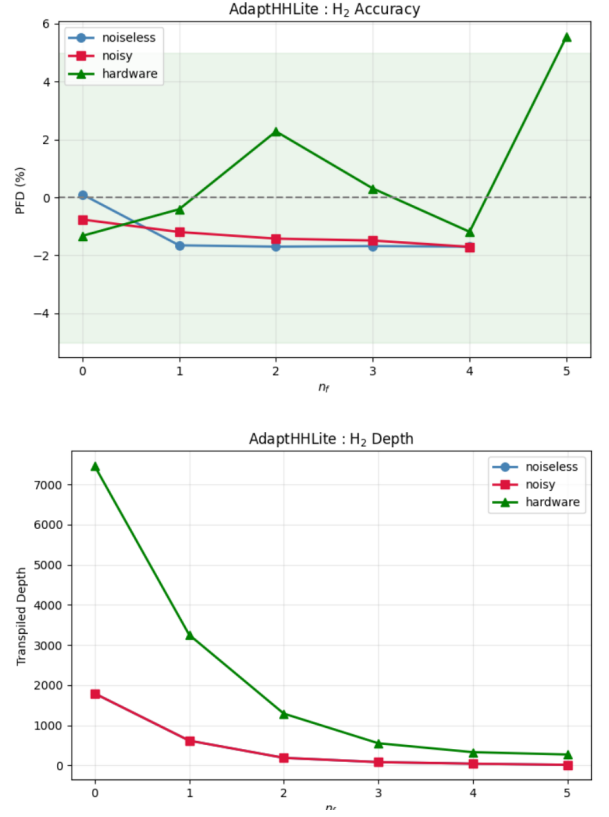


FIG. 5. AdaptHHLite performance for the H_2 molecular system.

The complete numerical results for all noiseless, noisy, and hardware implementations are provided in Appendix C3.

Several important observations may be made from the AdaptHHLite results:

- **Strong depth reduction:**

The transpiled circuit depth reduced from nearly **1800** to below **50** after fixing four clock qubits.

- **Effective qubit fixing:**

Unlike standard HHLite, the AdaptHHLite rescaling produced strongly biased QPE probabilities, allowing almost all clock qubits to be fixed successfully.

- **Large CX reduction:**

In the noiseless implementation, the entangling gate count decreased from **3053** CZ gates to only **86** CZ gates.

- **Hardware compression:**

On IBM hardware, the transpiled depth reduced from approximately **7500** to nearly **300**, while the CZ gate count dropped from over **3000** to below **150**.

- **Good energy accuracy:**

Despite the aggressive circuit compression, the estimated correlation energies remained close to the classical values across most implementations.

- **Breakdown at $n_f = 5$:**

When all clock qubits were fixed, the ancilla post-selection probability became extremely small, leading to unstable overlap estimation and larger deviations from the classical result.

2. *AdaptHHLite Results for LiH*

The AdaptHHLite framework was next benchmarked using the rescaled LiH molecular Hamiltonian:

- **Minimum diagonal estimate:** $d_{\min} = 0.500000$
- **Scaling factor:** $s = 0.062500$
- **Controlled rotation constant:** $C = 0.031250$
- **Scaled eigenvalues:** $[0.07572, 0.02952, 0.04346, 0.04019]$
- **QPE clock-qubit probabilities:** $P(1) = [0.110, 0.061, 0.050, 0.044, 0.044]$
- **Qubit fixing order:** $[4, 3, 2, 1, 0]$

Unlike the standard HHLite implementation for LiH, the AdaptHHLite rescaling procedure produced strongly biased QPE probabilities, enabling effective multi-qubit fixing across nearly the entire clock register.

The complete numerical results for all noiseless, noisy, and hardware implementations are provided in Appendix C 4.

The following observations can be made for this case:

- **Successful AdaptHHLite compression:**

Unlike standard HHLite, the AdaptHHLite rescaling procedure successfully enabled aggressive qubit fixing for the LiH system by producing strongly biased QPE probabilities.

- **Major depth reduction:**

The transpiled circuit depth reduced from nearly **1800** to below **50** after fixing four clock qubits.

- **Large reduction in entangling gates:**

In the hardware implementation, the entangling gate count decreased from **1300** CX gates to merely **22** CX gates.

- **Strong hardware compression:**

For IBM hardware executions, the transpiled depth reduced from approximately **7600** to nearly **400**, while the CZ gate count dropped from over **3000** to nearly **140**.

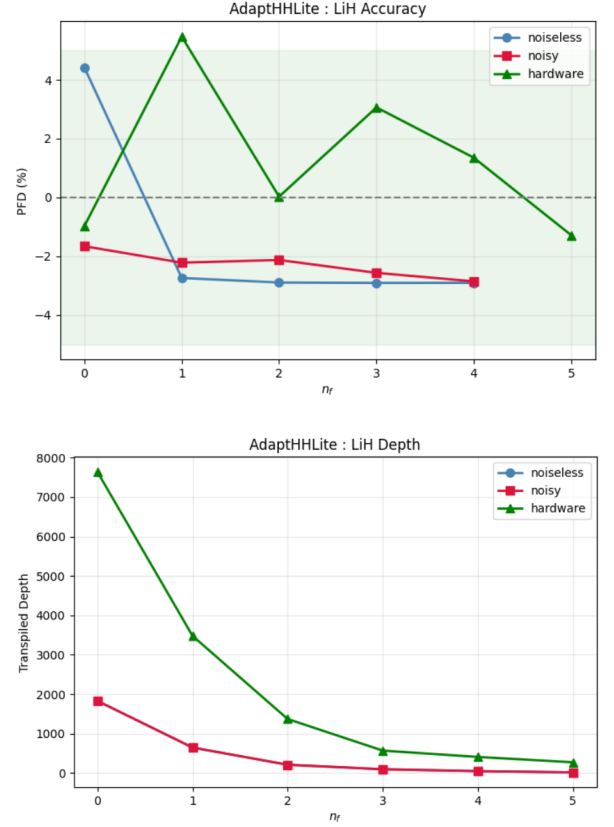


FIG. 6. AdaptHHLite performance for the LiH molecular system.

III. CLOSING REMARKS

A. Complexity of AdaptHHLite

The principal objective of the AdaptHHLite framework is the reduction of the circuit complexity associated with the standard HHL algorithm while preserving sufficient accuracy for the estimation of physically relevant observables such as the correlation energy.

For a sparse matrix of dimension $N \times N$, the complexity of the original HHL algorithm scales approximately as

$$\mathcal{O}(\log(N) s^2 \kappa^2 / \epsilon),$$

where s denotes the sparsity of the matrix, κ is the condition number, and ϵ represents the precision associated with the phase estimation procedure.

Within the AdaptHHLite framework, the matrix is first rescaled according to

$$A_{\text{sc}} = sA,$$

with the scaling chosen such that the smallest eigenvalue becomes approximately representable within the QPE register. As a consequence, the evolution time parameter scales as

$$t = \mathcal{O}(\kappa/\epsilon),$$

rather than the larger values typically required within standard HHL implementations.

Including the amplitude amplification overhead, the resulting complexity of AdaptHHLite becomes approximately

$$\mathcal{O}(\log(N) s^2 \kappa d_{\min}/\epsilon^2),$$

where $d_{\min}$ denotes the smallest diagonal element of the matrix. Since typically

$$d_{\min} < \kappa,$$

the AdaptHHLite framework behaves effectively as a sub- κ algorithm relative to the original HHL scaling.

Most importantly, however, the practical advantages of AdaptHHLite arise not solely from asymptotic complexity improvements, but from the dramatic reduction in circuit depth and entangling gate count achieved through the multi-qubit fixing protocol. By aggressively simplifying the controlled unitary evolutions and controlled rotation networks, the algorithm becomes substantially more amenable to implementation on present-day noisy intermediate-scale quantum (NISQ) hardware.

B. Conclusion

In this work, the HHL, HHLite, and AdaptHHLite algorithms proposed by the authors were implemented and benchmarked for reduced molecular Hamiltonians corresponding to the H_2 and LiH systems. The connection between linearised coupled cluster theory and quantum linear systems algorithms was first established, enabling the correlation energy problem to be reformulated as the solution of a linear system of equations.

The Hong–Ou–Mandel overlap estimation module employed within the original framework was also analysed, and its role in estimating the quantity

$$\langle b|x \rangle$$

without requiring complete reconstruction of the solution vector was discussed.

The standard HHL algorithm was observed to produce extremely deep circuits containing large numbers of entangling operations, making direct implementation challenging on contemporary hardware. The HHLite framework partially addressed this issue through the multi-qubit fixing protocol, yielding substantial circuit compression for systems whose QPE probability distributions were sufficiently biased. However, for systems such as LiH, the clock-qubit probabilities remained too close to 0.5, preventing effective qubit fixing and limiting the achievable reductions in circuit complexity.

The AdaptHHLite framework successfully addressed this limitation through the introduction of a physically motivated matrix rescaling procedure. By compressing

the eigenvalue spectrum into a range more naturally representable within the QPE register, strongly deterministic clock-qubit probability distributions were generated, enabling aggressive multi-qubit fixing even for the LiH system.

Across both H_2 and LiH, substantial reductions in transpiled circuit depth and entangling gate counts were observed for noiseless simulation, noisy simulation, and IBM hardware executions. In several cases, transpiled depths were reduced by more than an order of magnitude while still preserving reasonably accurate estimates of the molecular correlation energies.

These results demonstrate that physically motivated algorithmic simplifications may significantly improve the practical implementability of quantum linear systems algorithms on NISQ-era hardware. While large-scale fault-tolerant implementations of HHL remain technologically demanding, approaches such as HHLite and AdaptHHLite illustrate that carefully exploiting problem structure, approximate observables, and hardware-aware simplifications may already enable useful quantum chemistry calculations within near-term quantum computing architectures.

C. References

1. N. Baskaran, A. S. Rawat, A. Jayashankar, *et al.*, “Adapting the Harrow–Hassidim–Lloyd Algorithm to Quantum Many-Body Theory,” *Physical Review Research* **5**, 043113 (2023). DOI: 10.1103/PhysRevResearch.5.043113.
2. A. W. Harrow, A. Hassidim, and S. Lloyd, “Quantum Algorithm for Linear Systems of Equations,” *Physical Review Letters* **103**, 150502 (2009). DOI: 10.1103/PhysRevLett.103.150502.

Appendix A: Traditional Coupled Cluster Theory

In this appendix, we present a brief derivation of the linearised coupled cluster (LCC) framework, following the treatment of Ref. [?]. We begin from the CC ansatz, recast the LCC equations as a system of linear equations, and arrive at the expression for the correlation energy used in the main text.

Substituting the CC wavefunction $|\Psi\rangle = e^{\hat{T}}|\Phi_0\rangle$ into the Schrödinger equation gives

$$\hat{H}e^{\hat{T}}|\Phi_0\rangle = Ee^{\hat{T}}|\Phi_0\rangle. \quad (\text{A1})$$

Writing the Hamiltonian in normal-ordered form with respect to the Hartree–Fock function, one may subtract the vacuum energy to obtain

$$\{\hat{H}\}e^{\hat{T}}|\Phi_0\rangle = E_{\text{corr}}e^{\hat{T}}|\Phi_0\rangle, \quad (\text{A2})$$

where E_{corr} denotes the electron correlation energy and $\{\hat{H}\}$ denotes the normal-ordered portion of the Hamiltonian. Left-multiplying both sides by $e^{-\hat{T}}$ gives

$$e^{-\hat{T}}\{\hat{H}\}e^{\hat{T}}|\Phi_0\rangle = E_{\text{corr}}|\Phi_0\rangle, \quad (\text{A3})$$

$$\{\bar{H}\}|\Phi_0\rangle = E_{\text{corr}}|\Phi_0\rangle, \quad (\text{A4})$$

where $\bar{H}$ is the similarity-transformed Hamiltonian, expressed via the Baker–Campbell–Hausdorff (BCH) expansion as

$$\bar{H} = \hat{H} + [\hat{H}, \hat{T}] + \frac{1}{2!}[[\hat{H}, \hat{T}], \hat{T}] + \frac{1}{3!}[[[\hat{H}, \hat{T}], \hat{T}], \hat{T}] + \dots \quad (\text{A5})$$

The commutators are evaluated using Wick's theorem. Since the Hamiltonian consists of up to two-body terms, the BCH expansion terminates at the quartic power of $\hat{T}$.

Continuing from Eq. (A4) in the LCC approximation, the LCC equations are

$$\{(\hat{H} + \widehat{HT})\}|\Phi_0\rangle = E_{\text{corr}}|\Phi_0\rangle. \quad (\text{A6})$$

Writing the operators in terms of composite indices $P, Q, \dots$ representing orbital substitutions, and left-projecting by the excited functions $\langle\chi_P|$, where $|\chi_P\rangle = \{\hat{e}_P\}|\Phi_0\rangle$, one obtains

$$\langle\chi_P|\hat{h}|\Phi_0\rangle + \sum_Q \langle\chi_P|\hat{h}\{\hat{e}_Q\}|\Phi_0\rangle t_Q = 0, \quad \forall P, \quad (\text{A7})$$

which rearranges to

$$\sum_Q \langle\chi_P|\hat{h}|\chi_Q\rangle t_Q = -\langle\chi_P|\hat{h}|\Phi_0\rangle. \quad (\text{A8})$$

This is identified as the linear system $\mathbf{Ax} = -\mathbf{b}$ discussed in the main text. Finally, left-projecting Eq. (A6) by $\langle\Phi_0|$ yields the correlation energy

$$\langle\Phi_0|\hat{h}|\chi_P\rangle t_P = E_{\text{corr}}, \quad (\text{A9})$$

which, upon identifying the left-hand side as the inner product $-\mathbf{b}^\dagger A^{-1} \mathbf{b}$, gives

$$E_{\text{corr}} = -\mathbf{b}^\dagger A^{-1} \mathbf{b}, \quad (\text{A10})$$

consistent with Eq. (3) of the main text.

Appendix B: Derivation of the Hong–Ou–Mandel Overlap Formula

In the AdaptHHL framework, the correlation energy is determined through the overlap between the states $|b\rangle$ and

$$|x\rangle \propto A^{-1}|b\rangle.$$

The Hong–Ou–Mandel (HOM) module provides an interference-based procedure for estimating this overlap directly from measurement probabilities.

Consider two n -qubit states written in the computational basis as

$$|b\rangle = \sum_i b_i |i\rangle,$$

and

$$|x\rangle = \sum_j x_j |j\rangle,$$

where $i, j \in \{0, 1\}^n$.

The combined state is therefore

$$|b\rangle|x\rangle = \sum_{i,j} b_i x_j |i\rangle|j\rangle.$$

The HOM module first applies bitwise CNOT operations between the two registers. Taking the first register as the control and the second as the target gives the transformation

$$|i\rangle|j\rangle \longrightarrow |i\rangle|i \oplus j\rangle,$$

where $\oplus$ denotes bitwise XOR.

The state after the CNOT layer is therefore

$$|\Psi\rangle = \sum_{i,j} b_i x_j |i\rangle|i \oplus j\rangle.$$

Defining

$$\beta = i \oplus j,$$

so that

$$j = i \oplus \beta,$$

the state may equivalently be written as

$$|\Psi\rangle = \sum_{i,\beta} b_i x_{i \oplus \beta} |i\rangle|\beta\rangle.$$

Hadamard gates are then applied to every qubit in the first register. Using the identity

$$H^{\otimes n}|i\rangle = \frac{1}{\sqrt{2^n}} \sum_{\alpha} (-1)^{\alpha \cdot i} |\alpha\rangle,$$

where $\alpha \cdot i$ denotes the bitwise inner product modulo two, one obtains

$$|\Psi'\rangle = \frac{1}{\sqrt{2^n}} \sum_{\alpha,\beta} \left(\sum_i (-1)^{\alpha \cdot i} b_i x_{i \oplus \beta} \right) |\alpha\rangle|\beta\rangle.$$

The probability amplitude corresponding to the measurement outcome (α, β) is therefore

$$A_{\alpha\beta} = \frac{1}{\sqrt{2^n}} \sum_i (-1)^{\alpha \cdot i} b_i x_{i \oplus \beta},$$

and the corresponding measurement probability is

$$P(\alpha\beta) = |A_{\alpha\beta}|^2.$$

Consider now the parity-weighted summation

$$S = \sum_{\alpha,\beta} (-1)^{\alpha\cdot\beta} P(\alpha\beta).$$

Substituting the probability amplitudes gives

$$S = \frac{1}{2^n} \sum_{\alpha,\beta} (-1)^{\alpha\cdot\beta} \sum_{i,k} (-1)^{\alpha\cdot(i\oplus k)} b_i b_k^* x_{i\oplus\beta} x_{k\oplus\beta}^*.$$

Rearranging the summations yields

$$S = \frac{1}{2^n} \sum_{i,k} b_i b_k^* \sum_{\beta} x_{i\oplus\beta} x_{k\oplus\beta}^* \sum_{\alpha} (-1)^{\alpha\cdot(\beta\oplus i\oplus k)}.$$

The Walsh–Hadamard orthogonality relation

$$\sum_{\alpha} (-1)^{\alpha\cdot y} = 2^n \delta_{y,0}$$

then gives

$$\sum_{\alpha} (-1)^{\alpha\cdot(\beta\oplus i\oplus k)} = 2^n \delta_{\beta, i\oplus k}.$$

Consequently, only terms satisfying

$$\beta = i \oplus k$$

survive, leading to

$$S = \sum_{i,k} b_i b_k^* x_k x_i^*.$$

This expression factorises as

$$S = \left(\sum_i b_i x_i^* \right) \left(\sum_k b_k^* x_k \right),$$

which finally gives

$$|\langle b|x \rangle|^2 = \sum_{\alpha,\beta \in \{0,1\}^n} (-1)^{\alpha\cdot\beta} P(\alpha\beta). \quad (\text{B1})$$

The overlap between the two quantum states may therefore be extracted directly from experimentally accessible measurement probabilities through interference generated by the Hadamard transformations. This interference-based overlap estimation procedure forms the basis of the HOM module employed in AdaptHHL.

Appendix C: Detailed Numerical Results

1. HHLite Results for H_2

2. HHLite Results for LiH

3. AdaptHHLite Results for H_2

4. AdaptHHLite Results for LiH

TABLE I. Complete numerical results for the HHLite implementation applied to the H_2 molecular system.

Mode	n_f	Fixed Qubits	PFD (%)	Depth	Entangling Gates	Runtime (s)
Noiseless	0	0	+10.50	1795	1320 CX	0.99
	1	1	+0.51	687	464 CX	0.59
	2	2	-0.92	278	128 CX	0.23
	3	2	-0.85	278	128 CX	0.25
	4	2	-1.28	278	128 CX	0.21
	5	2	-1.00	278	128 CX	0.21
Noisy	0	0	-1.01	1795	1320 CX	51.06
	1	1	-0.96	687	464 CX	22.79
	2	2	-0.40	278	128 CX	7.37
	3	2	-0.41	278	128 CX	7.42
	4	2	-0.43	278	128 CX	7.46
	5	2	-0.86	278	128 CX	7.46
IBM Fez	0	0	-1.19	7449	3047 CZ	9.54
	1	1	-1.24	3317	1297 CZ	8.42
	2	2	-1.09	1619	608 CZ	8.60
	3	2	-0.76	1711	551 CZ	7.44
	4	2	-0.98	1645	575 CZ	8.21
	5	2	-0.56	1745	644 CZ	8.69

TABLE II. Complete numerical results for the HHLite implementation applied to the LiH molecular system.

Mode	n_f	Fixed Qubits	PFD (%)	Depth	Entangling Gates	Runtime (s)
Noiseless	0	0	-1.71	1831	1340 CX	1.25
	1	0	-0.50	1831	1340 CX	1.16
	2	0	-1.35	1831	1340 CX	1.25
	3	0	-1.00	1831	1340 CX	1.16
	4	0	-1.58	1831	1340 CX	1.16
	5	0	-1.62	1831	1340 CX	1.15
Noisy	0	0	-1.64	1831	1340 CX	45.69
	1	0	-1.24	1831	1340 CX	59.23
	2	0	-1.61	1831	1340 CX	55.62
	3	0	-1.13	1831	1340 CX	52.59
	4	0	-1.35	1831	1340 CX	59.15
	5	0	-1.74	1831	1340 CX	59.31
IBM Fez	0	0	-1.49	7614	3097 CZ	21.36
	1	0	-0.45	7646	3109 CZ	315.94
	2	0	-0.66	7590	3070 CZ	10.39
	3	0	-1.20	9110	3772 CZ	10.56
	4	0	-1.16	7590	3055 CZ	9.45
	5	0	-0.07	7602	3076 CZ	11.23

TABLE III. Complete numerical results for the AdaptHHLite implementation applied to the H_2 molecular system.

Mode	n_f	Fixed Qubits	PFD (%)	Depth	Entangling Gates	Runtime (s)
Noiseless	0	0	+0.10	1795	1320 CX	—
	1	1	-1.65	620	424 CX	—
	2	2	-1.70	191	90 CX	—
	3	3	-1.68	85	38 CX	—
	4	4	-1.70	45	18 CX	—
	5	5	NaN	18	0 CX	—
Noisy	0	0	-0.76	1795	1320 CX	—
	1	1	-1.19	620	424 CX	—
	2	2	-1.42	191	90 CX	—
	3	3	-1.49	85	38 CX	—
	4	4	-1.71	45	18 CX	—
	5	5	NaN	18	0 CX	—
IBM Fez	0	0	-1.33	7463	3053 CZ	—
	1	1	-0.41	3255	1140 CZ	—
	2	2	+2.28	1293	472 CZ	—
	3	3	+0.31	554	199 CZ	—
	4	4	-1.19	332	137 CZ	—
	5	5	+5.55	274	86 CZ	—

TABLE IV. Complete numerical results for the AdaptHHLite implementation applied to the LiH molecular system.

Mode	n_f	Fixed Qubits	PFD (%)	Depth	Entangling Gates	Runtime (s)
Noiseless	0	0	+4.42	1831	1340 CX	—
	1	1	-2.74	648	440 CX	—
	2	2	-2.90	211	102 CX	—
	3	3	-2.91	97	46 CX	—
	4	4	-2.91	49	22 CX	—
	5	5	NaN	18	0 CX	—
Noisy	0	0	-1.66	1831	1340 CX	—
	1	1	-2.22	648	440 CX	—
	2	2	-2.13	211	102 CX	—
	3	3	-2.57	97	46 CX	—
	4	4	-2.86	49	22 CX	—
	5	5	NaN	18	0 CX	—
IBM Fez	0	0	-0.98	7640	3100 CZ	—
	1	1	+5.47	3480	1150 CZ	—
	2	2	+0.02	1371	433 CZ	—
	3	3	+3.06	570	201 CZ	—
	4	4	+1.35	409	138 CZ	—
	5	5	-1.29	274	86 CZ	—